Hironmoy Roy Rudra

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Education

BRAC University — Dhaka, Bangladesh
BSc in Computer Science, CGPA 3.65/4.00 — Sep 2022 – Jan 2027 (expected)

- **Research interests:** robotics, autonomous systems, high-fidelity simulation, perception, IoT.
- **Honors:** VC's List for Academic Excellence (4 semesters); Dean's List for Academic Excellence (3 semesters).
- **Thesis:** TSDLoRA: Prioritizing Task-Specific Layers for Efficient Low-Rank Adaptation in Encoder-Only Transformers (2026).

National College of Education — Narsingdi, Bangladesh
Higher Secondary Certificate, Science group — 2019 – 2021

Research Interests

My research asks how a robot can act safely when its own sensing is uncertain, and how simulation can be made trustworthy enough to answer it. First, uncertainty-aware perception and navigation for field and planetary rovers: certified terrain traversal under sensing uncertainty, classical navigation stacks on hidden-world benchmark suites, and field rover autonomy (CERTUS, BARN, Mongol-Tori). Second, event-based and neuromorphic vision for agile multi-robot teams, where frame cameras struggle under hard ego-motion, motion blur and rapidly changing illumination (COVE, evs-bench, hifienv). Third, datasets and benchmarks for dense, unstructured traffic in the Global South (DhakaSense). The tie is simulation that can be trusted as ground truth: label-correct, sensor-clean environments in which perception and navigation are measured before deployment on real robots.

Publications and Manuscripts

- **Matched Controls for Probe-Based LoRA Rank Allocation: A Negative Result for One Allocator under a Rank-Coupled Initialization**. *Under review, Transactions on Machine Learning Research (submitted 2026) (first author).* Compares TSDLoRA with a budget-matched uniform-rank control, LoRA and GoRA over five seeds on eight commonsense tasks, three decoders and 11 of 24 GLUE cells; TSDLoRA trails the control in 26 of 35 cells.
- **DhakaSense: A Multimodal Driving Dataset and Benchmark for Dense, Heterogeneous Traffic in Dhaka**. *Under review, IEEE ICRA 2027 (submitted 2026) (second author).* Presents 14 h 35 min of synchronized LiDAR, stereo-depth, six-camera and GNSS/INS driving data over 175 Dhaka scenes, labelled semi-automatically across an 8-class taxonomy and benchmarked per traffic-density level.
- **CERTUS: Certified Terrain Traversal Under Sensing Uncertainty, with Certification-Driven Active Perception**. *In preparation, target IEEE/RSJ IROS 2027 (first author).* Gates traversal on step, slope and clearance with a distribution-free (Cantelli) margin, predicts which depth sensor can certify which rover, and resolves blocked cells by active perception; validated in Isaac Sim before deployment on a real rover.
- **COVE-Sim: counterfactual, recipient-specific communication for event-camera UAV teams (working title)**. *In preparation, target IROS 2027 (CoRL 2027 fallback) (first author).* Decides what each event-camera UAV tells which teammate from the counterfactual value of a message to that recipient's task loss, giving the first collaborative-perception system with an event-derived shared representation; simulation study in Cosys-AirSim.
- **COVE multi-UAV hazard detection and avoidance dataset (working title)**. *In preparation, target NeurIPS 2027 Datasets & Benchmarks track.* Covers five industrial environment families (warehouse, factory, construction, port, mine) with seeded layouts, exact label geometry and QC gates, hazards at rule-breaking speeds, and event and RGB capture in Cosys-AirSim.

Research Experience

BRACU Mongol-Tori (University Mars Rover Team) — Dhaka, Bangladesh
University Project Team – multiple roles, ranked 8th at University Rover Challenge 2025 — Nov 2023 – Present

- **Co-Advisor (Nov 2025 – Present):** Direct technical strategy for the European Rover Challenge build. Mentor undergraduate engineers on autonomous navigation, perception, ROS 2 architecture, and sim-to-real validation, and push the team to write up competition work to publication standards.
- **Sub-Team Lead, Autonomous (Sep 2024 – Aug 2025):** Led the autonomy sub-team. Built and deployed the autonomy stack, SLAM pipeline, and path-planning algorithms for the rover. Ran the autonomous missions on-site at the University Rover Challenge in Utah and managed live telemetry during competition runs.
- **Control Systems Engineer (Feb 2024 – Aug 2024):** Wrote the ROS 2 communication and control architecture from scratch. Implemented inverse kinematics for a 3-DOF arm using ArUco markers and an OpenCV pipeline. Worked on camera calibration for the rover's stereo and RGB feeds.
- **Contributor (Nov 2023 – Feb 2024):** Studied rover control on Arduino and Raspberry Pi, and assessed the feasibility of moving the rover stack onto ROS.

Kibo Robot Programming Challenge, 6th Edition — Dhaka, Bangladesh
6th Place, National Round, JAXA – Computer Vision and Autonomous Sign Detection — Aug 2025 – Feb 2026

- Placed 6th in the national round with a real-time CV pipeline for autonomous sign detection in JAXA's microgravity simulator.
- Wrote real-time detection and classification for visual markers, tuned the CV models for speed and accuracy under simulator constraints, and connected the perception module to the navigation stack for autonomous targeting.

Research Projects

- **CERTUS navigation stack: uncertainty-aware terrain traversal for rovers** *(Sep 2026 – Present)*: Built the sim-first research stack behind CERTUS: a terrain pipeline from ZED 2i stereo depth through `elevation_mapping_cupy` to slope, roughness and uncertainty

layers, with TOPP-RA (cvxpy) velocity profiles whose limits vary along the path; Isaac Sim 6.0.1 scenes for Clearpath Husky, Jackal and Warthog (URDF to USD) and a ROS 2 Jazzy distrobox container. Wrote an idempotent, staged `setup.sh` that fetches every non-source dependency at a pinned version checked by MD5, sha256 or commit, and a `verify` stage that re-runs the tests and a headless Husky drive and fails unless it reproduces the reported run.

- **BARN Challenge 2027 preparation** *(Jul 2026 – Aug 2026)*: Built a classical navigation stack in C++ on ROS 2 Jazzy for a Clearpath Jackal with 2-D LiDAR: log-odds occupancy grid with Euclidean distance transform, two-phase Dijkstra and lattice A*, elastic-band reference, sequentially-linearised MPC solved as a QP with OSQP at 30 Hz, swept-footprint safety shield and backtracking recovery. Scripted a scored campaign over the BARN 2026 public suite (50 worlds × 10 trials, captured manifest, two metric reports) and wrote a nine-chapter tutorial and a seven-demo interactive companion site.
- **hifi-environment-builder** *(Oct 2026)*: Wrote a Python toolkit (`hifienv`) and Claude Code skill that build label-correct, sensor-clean, licence-clean, reproducible simulation environments for Unreal Engine 5/Cosys-AirSim and Isaac Sim 6 (USD). Fetches CC0/CC-BY PBR materials from Poly Haven, ambientCG and MatSynth with per-file provenance (source, licence, sha256), builds GLB kits headless in Blender, generates seeded layouts, imports into Unreal or writes collider and visual USD, and gates every property: licence allowlist, label parity, null tests, 1 mm collider parity; 289 tests, MIT.
- **evs-bench (private repository)** *(Jul 2026 – Sep 2026)*: Built Python benchmark tooling for event-based detection of high-speed targets: a dataset contract with validator and QC reports, corrected ESIM event synthesis (persistent per-pixel reference levels, log-intensity interpolation, refractory period) pinned by analytic unit tests, class-agnostic RGB and event YOLO11 baselines, a dt-aware Kalman/SORT tracker, and a time-to-first-detection metric with k-consecutive confirmation and oracle reference curves. Added AirSim fly-and-record capture that replays a logged trajectory through a stepped 2 ms pipeline for 500 Hz output; 57 tests, MIT.
- **Closed-loop rocker-differential rover simulation – Rudra-roy/rocker-loop-sim** *(Sep 2026)*: Built the closed kinematic loop of a rocker-differential rover from Fusion 360 CAD in PyBullet and Gazebo Classic (ROS 2 Humble): a URDF tree plus point-to-point constraints in PyBullet, native ball joints in Gazebo, whose world is Apollo 15 terrain with seeded rocks. Verified loop closure to 0.003 mm (PyBullet) and 0.05 mm (Gazebo) driving on 16° slopes, found dead points at $-21.5°$ and $+40°$ of rocker travel, and recommended $\pm 15°$ hard stops. Compared skid steering with $\pm 90°$ wheel steering.
- **BMHS path planner – Rudra-roy/BHMS-Kinematics-aware-path-planner** *(2026)*: Wrote a bidirectional multi-heuristic hybrid A* planner for Ackermann and differential-drive vehicles that grows forward and backward trees from motion primitives, guided by two obstacle-aware 2-D Dijkstra heuristics computed in parallel; the C++17 core (pybind11) achieves a 100x+ speedup over native Python implementations. Simplifies paths with clearance-aware Bresenham line-of-sight checks and smooths Ackermann paths with quadratic Bézier curves that respect the minimum turning radius. Ported the core to an experimental ROS 2 Humble C++ package exposing ComputePathToPose and FollowPath actions.
- **Unity digital twin and ROS 2 autonomy workspace for BRACU Mongol-Tori – bracu-mongoltori/simple_robot_ws** *(Dec 2025 – Present)*: Built a digital twin of the Mars rover in Unity with a MuJoCo physics backend connected to ROS 2: rocker-bogie suspension from URDF joints and custom mesh collisions, a replicated sensor suite, and synthetic data for SLAM and LiDAR-GNSS fusion tests on simulated planetary terrain. Built the ROS 2 workspace that is the rover's autonomy backend and tied its control pipeline to the twin for software-in-the-loop testing, with modular path-planning and kinematics nodes to validate edge cases before hardware deployment.
- **3D LiDAR map-based topographical path planner** *(Aug 2025)*: Wrote a path planner based on artificial potential fields for safe autonomous waypoint traversal. Reads raw LiDAR point clouds to estimate terrain slope and isolate high-risk zones in real time. Outperformed BFS, DFS, Dijkstra, hill climbing and conditional A* on traversal efficiency and route safety on rough surfaces by using continuous spatial heuristics rather than discrete graph search.
- **DhakaSense capture and annotation stack** *(Jul 2026 – Sep 2026)*: Built the in-vehicle recording stack for a multimodal Dhaka driving dataset: supervisor and watchdog over ZED SVO2, RealSense and chunked-audio recorders, an SBG Ellipse-D INS as master clock, optical-clock sync validation, dual-copy SHA-256 offload and QC reports. Wrote a licence-free PPK chain (UBX to RINEX, RTKLIB), a CVAT label generator, and a remote export orchestrator that runs the nuScenes exporter over NFS, attaches PPK-enriched ego poses with fix status and sigmas, and seals exports with BLAKE2b manifests and a ledger.

Teaching and Mentoring

Robotics Club of BRAC University — Dhaka, Bangladesh
Student Instructor – ROS and physics simulation — Oct 2025 – Nov 2025

- Taught fundamental and intermediate ROS topics and physics simulation in Gazebo and Webots to undergraduate club members.
- Helped students debug code, structure their workspaces, and turn rough ideas into running simulated robot projects.

First Global Challenge (FGC) Team Bangladesh — Dhaka, Bangladesh
Technical Mentor – national robotics team — Sep 2024 – Feb 2025

- Spent six months training the national team on programming, problem decomposition, logic building, and competition strategy.
- Built an end-to-end rice leaf disease detection pipeline in PyTorch and optimized inference with TensorRT for high-speed edge deployment on the team's robot.

Honors and Awards

- 8th Place, University Rover Challenge 2025 – The Mars Society, Hanksville, Utah, USA (Jun 2025).
- 21st Place, University Rover Challenge 2024 – The Mars Society, Hanksville, Utah, USA (Jun 2024).
- 6th Place, National Round, Kibo Robot Programming Challenge 2025 – JAXA, Tokyo, Japan (Feb 2026).

Technical Skills

- **Programming & Databases:** Python (advanced), C/C++, C#, JavaScript, TypeScript, SQL, Bash
- **Deep Learning & Computer Vision:** PyTorch (advanced), TensorFlow, TensorRT, Keras, OpenCV, YOLO Ultralytics models, Tesseract OCR, classical computer vision
- **AI / ML / NLP:** Scikit-learn, RAG systems, vector databases (Pinecone, ChromaDB), LLMs (GPT, Llama, DeepSeek), Hugging Face Transformers, Reinforcement Learning (DQN, PPO), Optuna, LoRA
- **Robotics & Simulation:** ROS 2; physics simulation in Gazebo, Webots, PyBullet, MuJoCo; high-fidelity simulators including Isaac Sim, Unreal Engine 5/Cosys-AirSim, CARLA, Unity; path planning, trajectory optimisation (TOPP-RA, cvxpy), inverse kinematics, control systems design, sensor integration and calibration
- **Embedded Systems:** Arduino, Raspberry Pi, ESP32, STM32
- **Data & MLOps:** Pandas, NumPy, feature engineering, data augmentation, bioinformatics pipelines; Docker, Git/GitHub, CI/CD, Weights & Biases, model quantization
- **3D Design & Web:** Fusion 360, OnShape, Blender (basic); Flask, FastAPI, React Native (Expo Go), Tailwind CSS, Laravel, MySQL, MongoDB, SQLite, REST API design
- **Tools & Platforms:** Reverse engineering, GUI development (PyQt, Tkinter), Linux (Ubuntu, Raspbian), Jupyter notebooks

Certifications and Training

- Deep Learning Specialization – Coursera (2024).
- CS50: Computer Science – Harvard (2023).
- 100 Days of Code with Python – Udemy (2023).
- Data Structures and Algorithms with Python 3 – Udemy (2023).
- Ethical Hacking with zSecurity – Udemy (2023).
- Kibo RPC 6th Edition Participant – JAXA (Dec 2025).
- Duke of Edinburgh Bronze Award (2023).

References

Dr. Md. Khalilur Rhaman, PhD – Professor, CSE Department, BRAC University – khalilur@bracu.ac.bd
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